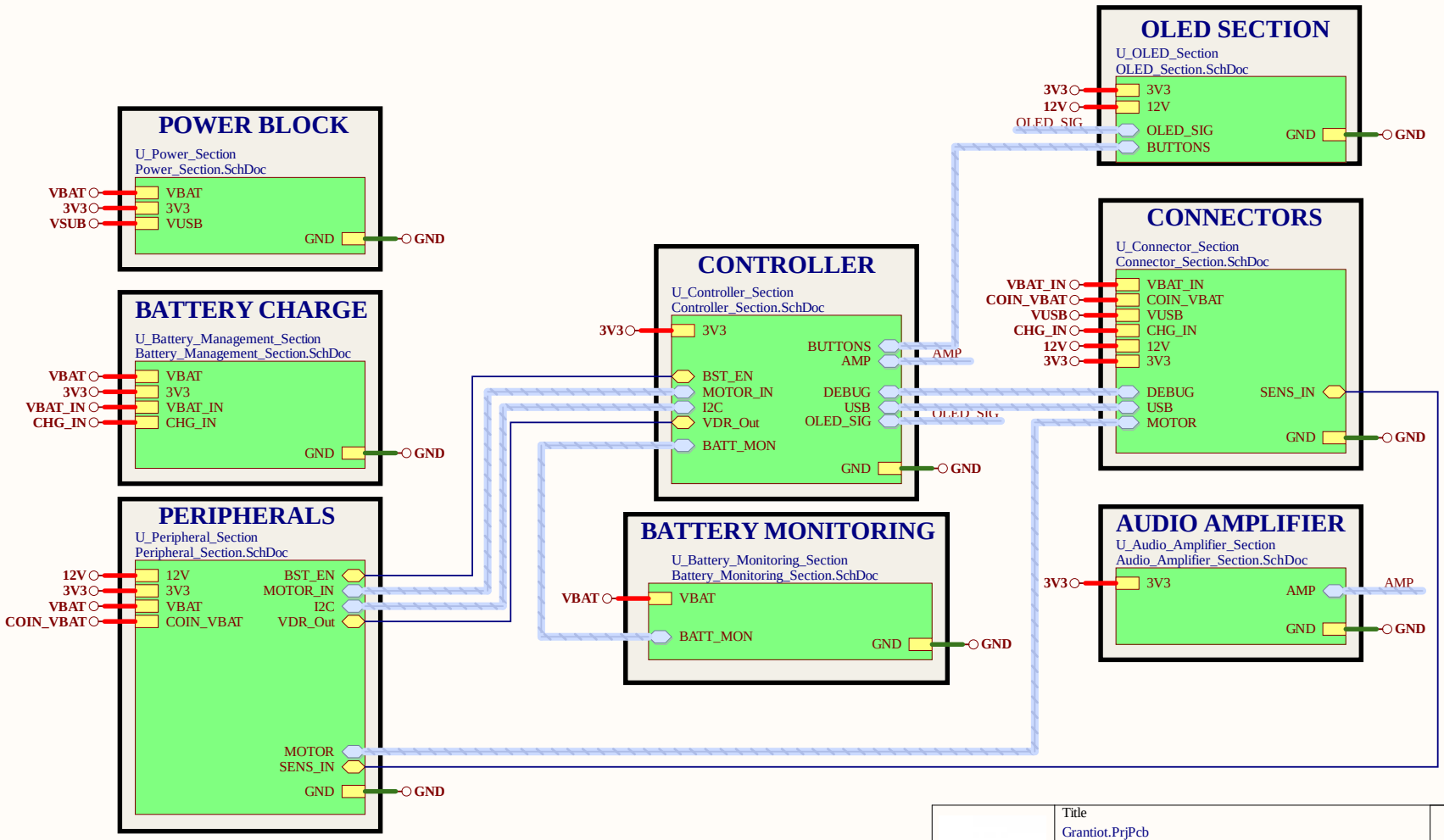
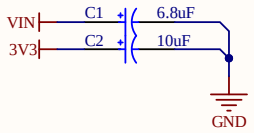
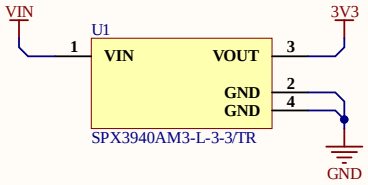


Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/21/2022	Sheet 1 of	9
File:	E:\Git\..\Top Sheet.SchDoc	Drawn By:	Huzaifa Ali

DECOUPLING CAPS



MCU VOLTAGE REGULATOR IC



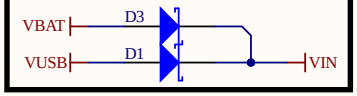
TEST POINT



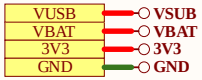
POWER INDICATION LED



ORING CIRCUIT

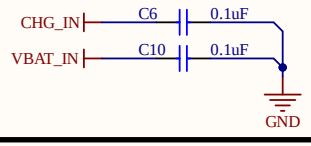


VBAT=6V
VUSB=5V
VF of Diode= 360mV @1A
VR=30V
IF=1A

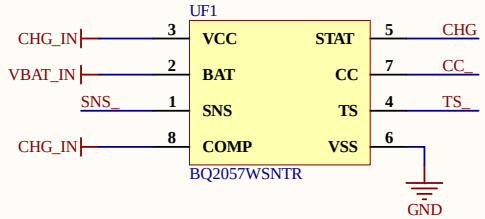


Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/21/2022		Sheet 2 of 9	
File: E:\Git\..\Power_Section.SchDoc		Drawn By: Huzaifa Ali	

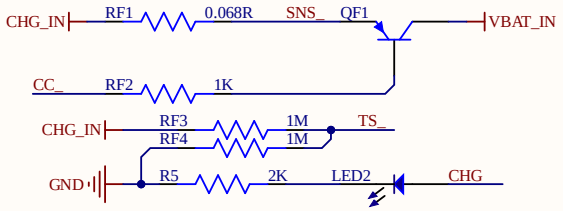
DECOUPLING CAPS



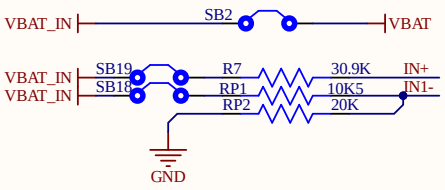
BATTERY CHARGING IC



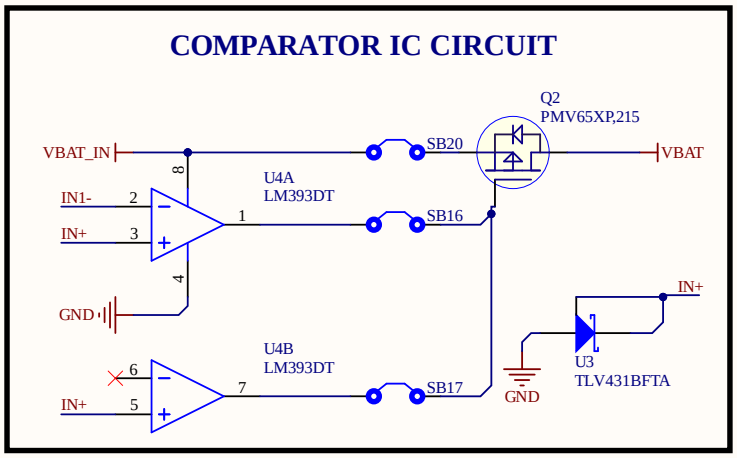
BATTERY CHARGING IC CIRCUIT



Divider for Battery Cutoff at 3.7V (3xAA Cell)



COMPARATOR IC CIRCUIT

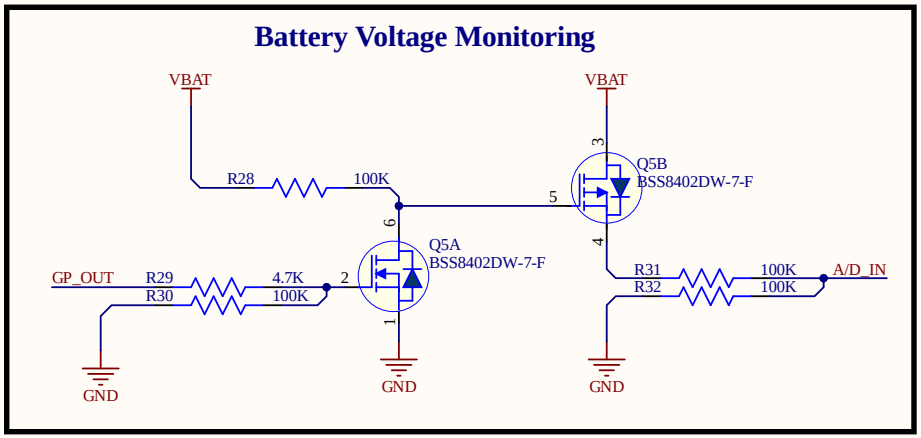


CHG_IN	**CHG_IN**
VBAT_IN	**VBAT_IN**
VBAT	**VBAT**
3V3	**3V3**
GND	**GND**

eSpark Consulting Group

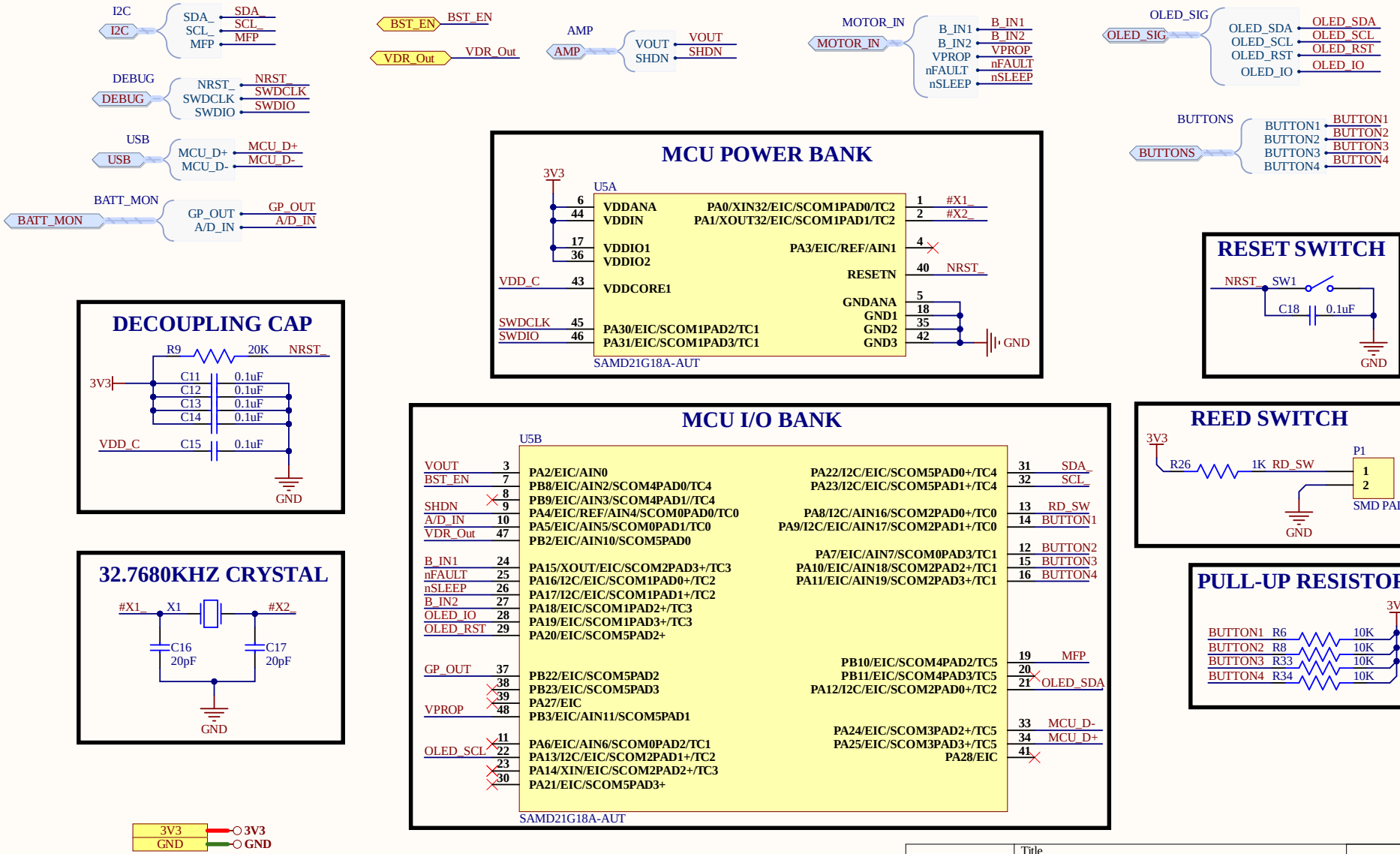
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/21/2022	Sheet	3 of 9
File:	E:\Git\..\Battery_Management_Section.SchDoc	Drawn By:	Huzaifa Ali





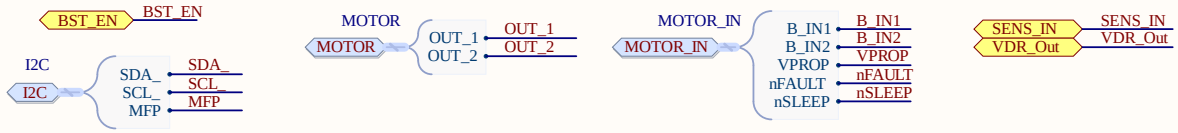


	Title: Grantiot.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 3/21/2022	Sheet 4 of 9	
	File: E:\Git\..\Battery_Monitoring_Section.SchDoc	Drawn By: Huzaifa Ali	

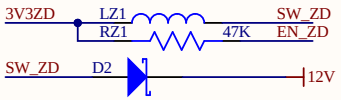


Controller Section

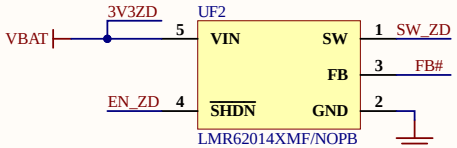
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/21/2022	Sheet 5 of	9
File:	E:\Git\..\Controller_Section.SchDoc	Drawn By:	Huzaifa Ali



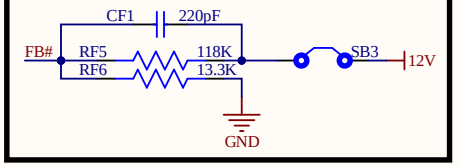
SUPPORTING CIRCUIT



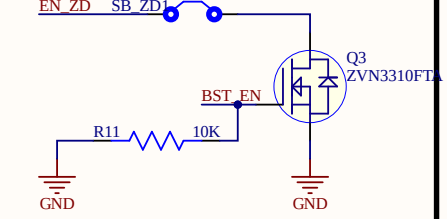
12V BOOST REGULATOR



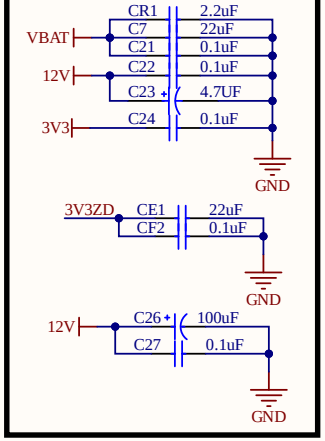
FEEDBACK NETWORK



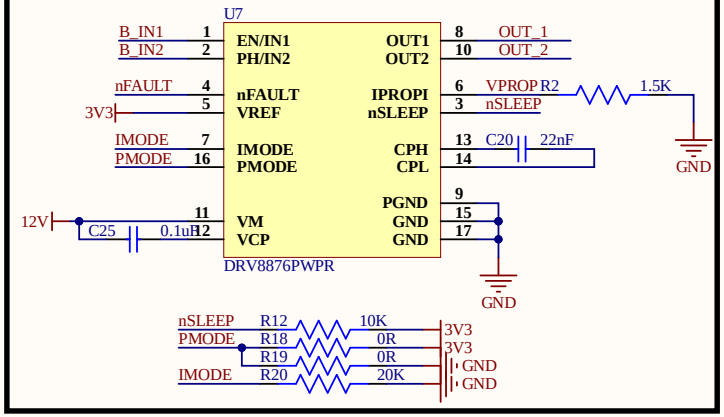
FEEDBACK NETWORK



DECOUPLING CAPS



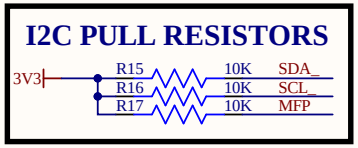
MOTOR DRIVER IC



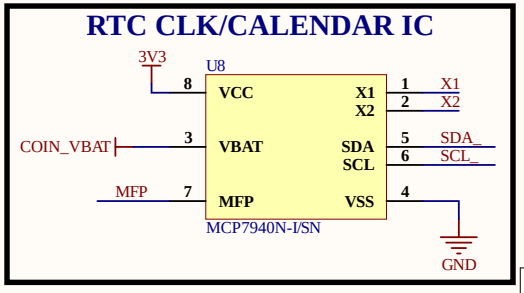
SENSOR INPUT CONVERSION



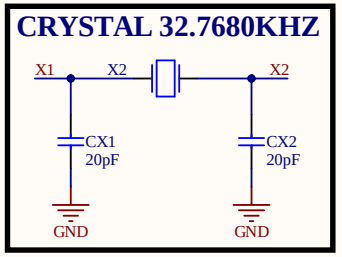
I2C PULL RESISTORS

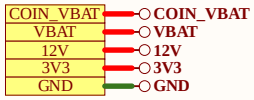


RTC CLK/CALENDAR IC



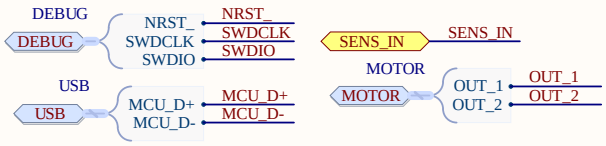
CRYSTAL 32.7680KHZ



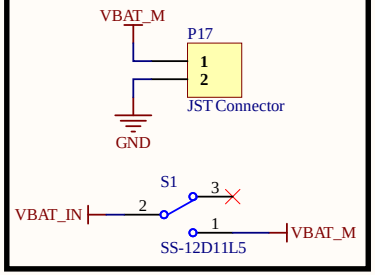


Peripheral Section

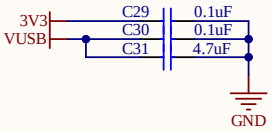
Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/21/2022		Sheet 6 of 9	
File: E:\Git\..\Peripheral_Section.SchDoc		Drawn By: Huzaifa Ali	



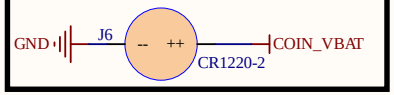
BATTERY INPUT



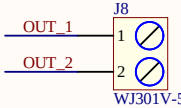
DECOUPLING CAP



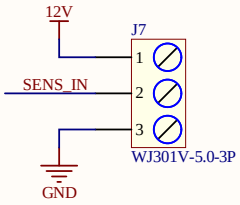
COIN CELL CONNECTOR



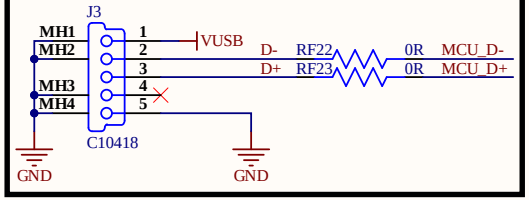
MOTOR OUTPUT



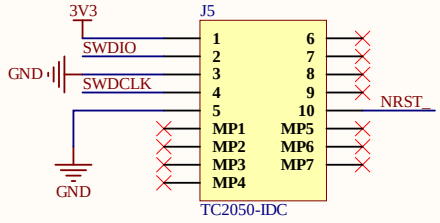
PROXIMITY SENSOR



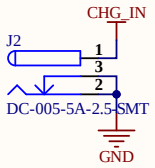
USB DATA TRANSFER



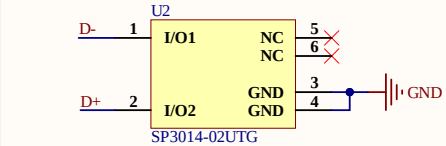
JTAG DEBUGGING CONNECTOR



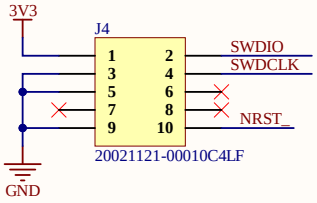
CHARGING ADAPTER



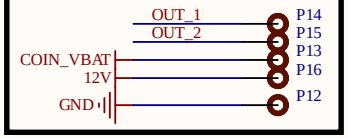
TVS ARRAY FOR USB INPUT

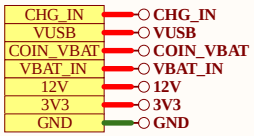


SERIAL WIRE DEBUGGING HEADER



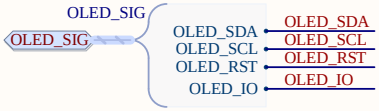
TEST POINTS



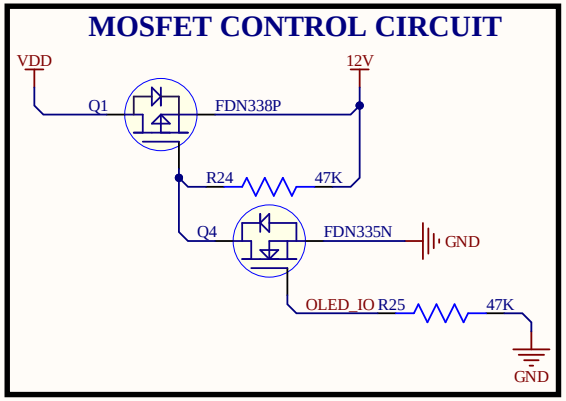


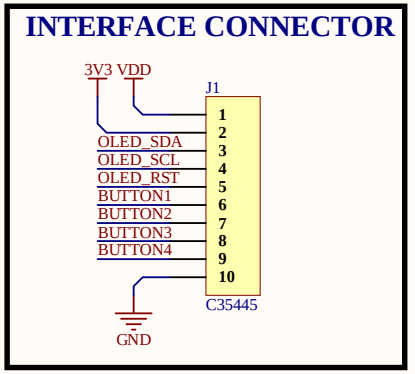
Connector Section

Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/21/2022		Sheet 7 of 9	
File: E:\Git\..\Connector_Section.SchDoc		Drawn By: Huzaifa Ali	











Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/21/2022		Sheet 8 of 9	
File: E:\Git\..\OLED_Section.SchDoc		Drawn By: Huzaifa Ali	

AMP

AMP

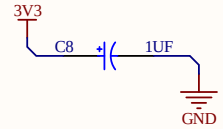
VOUT

SHDN

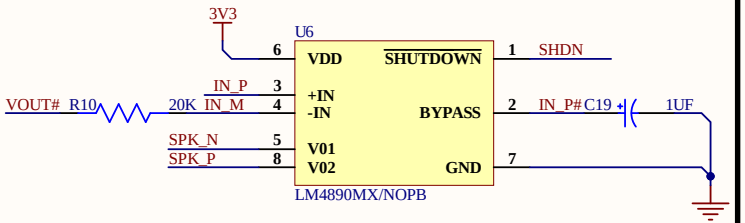
VOUT

SHDN

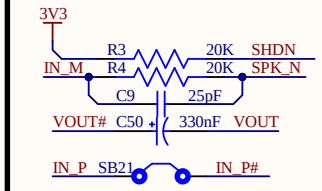
DECOUPLING CAP



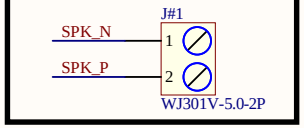
1W AUDIO AMPLIFIER IC



SUPPORTING CIRCUIT

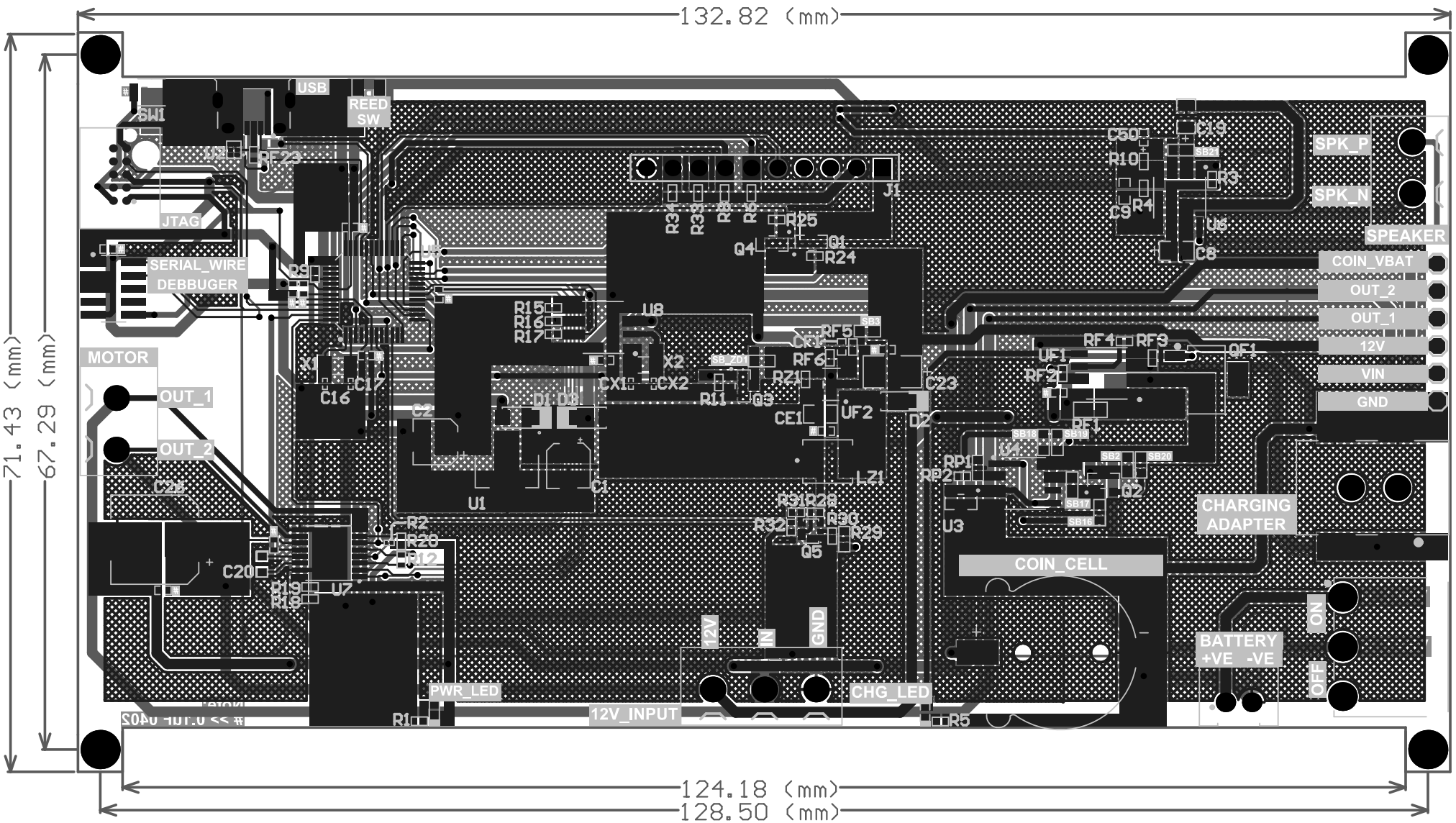


SPEAKER OUTPUT





Title	Grantiot.PrjPcb		
Size	A4	Number	1
Revision	1		
Date:	3/21/2022	Sheet	9 of 9
File:	E:\Git\..\Audio_Amplifier_Section.SchDoc	Drawn By:	Huzaifa Ali


132.82 (mm)
71.43 (mm)
67.29 (mm)
124.18 (mm)
128.50 (mm)
USB
REED SW
SW1
RF23
JTAG
SERIAL_WIRE DEBBUGER
MOTOR
OUT_1
OUT_2
J1
SPK_P
SPK_N
SPEAKER
COIN_VBAT
OUT_2
OUT_1
12V
VIN
GND
CHARGING ADAPTER
COIN_CELL
BATTERY +VE -VE
ON
OFF
PWR_LED
CHG_LED
12V_INPUT
12V
IN
GND
U1
U2
U3
U4
U5
U6
U7
U8
Q1
Q2
Q3
Q4
Q5
D1
D2
D3
C1
C2
C8
C9
C16
C17
C20
C23
C50
R1
R2
R3
R4
R5
R10
R11
R15
R16
R17
R19
R20
R24
R25
R28
R29
R30
R31
R32
RP1
RP2
RF1
RF2
RF3
RF4
RF5
RF6
RZ1
CE1
LZ1
UF1
UF2
X1
X2
CX1
CX2
SB1
SB2
SB3
SB4
SB16
SB17
SB18
SB19
SB20
SB21